

Problem

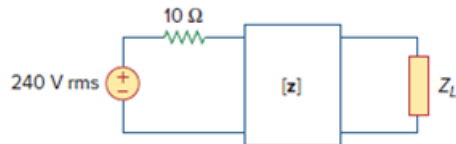
For the two-port in Fig. 19.75, show that

$$\frac{I_2}{I_1} = \frac{-g_{21}}{g_{11}Z_L + \Delta_g}$$

$$\frac{V_2}{V_1} = \frac{g_{21}Z_L}{(1 + g_{11}Z_s)(g_{22} + Z_L) - g_{21}g_{12}Z_s}$$

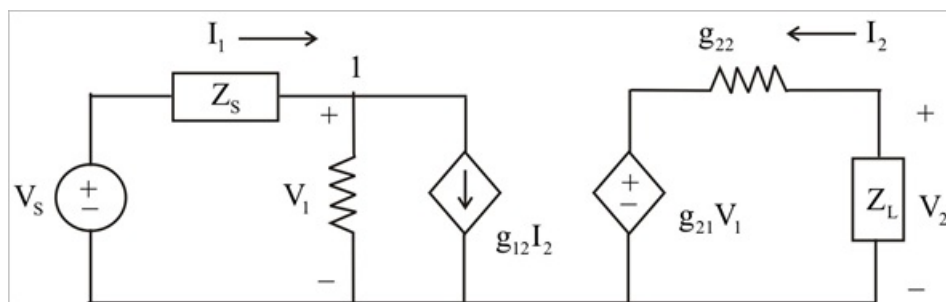
where Δ_g is the determinant of $[g]$ matrix.

Figure 19.75



Step-by-step solution

Step 1 of 5



Step 2 of 5

At node 1. $I_1 = g_{11}V_1 + g_{12}I_2$

$$\frac{I_1}{I_2} = g_{11} \frac{V_1}{I_2} + g_{12} \quad -(1)$$

$$V_2 = \frac{-g_{21}V_1}{g_{22} + Z_L} \times Z_L \quad -(2)$$

$$I_2 = \frac{-g_{21}V_1}{g_{22} + Z_L} \quad -(3a)$$

$$\frac{V_1}{I_2} = \frac{g_{22} + Z_L}{-g_{21}}$$

Step 3 of 5

substituting in (1) $\frac{I_1}{I_2} = \frac{g_{11}g_{22} + g_{11}Z_L}{-g_{21}} + g_{12}$

$$= \frac{g_{11}g_{22} + g_{11}Z_L - g_{12}g_{21}}{-g_{21}}$$

$$\boxed{\frac{I_1}{I_2} = \frac{-g_{21}}{g_{11}Z_L + \Delta g}}$$

where $\Delta g = g_{11}g_{22} - g_{12}g_{21}$

from (2) we have $\frac{V_2}{V_1} = \frac{-g_{21}Z_L}{g_{22} + Z_L} \quad -(4)$

Step 4 of 5

At node 1.

$$\frac{V_s - V_1}{Z_s} = V_1 g_{11} + g_{12} I_2$$

Dividing by V_1 and rearranging

$$\frac{V_s}{V_1} = 1 + g_{11}Z_s + g_{12}Z_s \frac{I_2}{V_1}$$

From 3a

$$\begin{aligned} \frac{V_s}{V_1} &= (1 + g_{11}Z_s) + g_{12}Z_s \left(\frac{-g_{21}}{g_{22} + Z_L} \right) \\ &= \frac{(1 + g_{11}Z_s)(g_{22} + Z_L) - g_{21}g_{12}Z_L}{g_{22} + Z_L} \end{aligned}$$

Step 5 of 5

From (4) $V_2 = \frac{-g_{21} + Z_L}{g_{22} + Z_L} V_1$

$$\frac{V_2}{V_s} = \frac{-g_{21} + Z_L}{g_{22} + Z_L} \times \frac{g_{22} + Z_L}{(1 + g_{11}Z_s)(g_{22} + Z_L) - g_{12}g_{21}Z_L}$$